COEFFICIENTS OF SYMMETRIC SQUARE L-FUNCTIONS

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ABSTRACT. Let $\lambda_{\mathrm{sym}^2 f}(n)$ be the nth coefficient in the Dirichlet series of the symmetric square L-function associated to a holomorphic primitive cusp form f. We prove $\Omega_\pm$ results for $\lambda_{\mathrm{sym}^2 f}(n)$ and evaluate the number of positive (resp. negative) $\lambda_{\mathrm{sym}^2 f}(n)$ in some intervals.

Dedicated to Professor Yuan Wang's 80th birthday

1. INTRODUCTION

A central problem in the study of (classical) modular forms is to unravel the behaviour of Hecke eigenvalues $\lambda_f(n)$, see for instance, [15, 14, 9, 13, 11] and the references therein. To some extent the knowledge is pleasantly built, however it is scanty for automorphic forms of higher rank groups. This question is plausibly difficult. Indeed, even the case of GL_3 is not well cultivated. As an initial step, we consider the special family of GL_3 forms that are the Gelbart-Jacquet lift of holomorphic GL_2 forms. More precisely we investigate the coefficients of the symmetric square L-function associated to a holomorphic primitive cusp form for congruence subgroups.

To be specific we fix the notation and describe the problems of our interest. Let $k \geqslant 2$ be an even integer and $N \geqslant 1$ be a squarefree integer. The (holomorphic) primitive cusp forms are defined as in [8, Section 14.7] and $\mathrm{H}_k^*(N)$ denotes the set of all primitive cusp forms of weight k and of level N. A feature of $f \in \mathrm{H}_k^*(N)$ is that its Fourier expansion at the cusp ∞,

$$f(z) = \sum_{n=1}^{\infty} \lambda_f(n) n^{(k-1)/2} \mathrm{e}^{2\pi \mathrm{i} n z} \qquad (\Im m z > 0),$$

where the nth normalized Fourier coefficient $\lambda_f(n)$ of f is the eigenvalue under the Hecke operator T_n. The following are nowadays widely known:

(1) $\lambda_f(n)$ is a real multiplicative function, and

$$\lambda_f(m)\lambda_f(n) = \sum_{\substack{d \mid (m,n) \\ (d,N)=1}} \lambda_f\left(\frac{mn}{d^2}\right) \tag{1.1}$$

for all integers $m \geqslant 1$ and $n \geqslant 1$.

(2) For any prime number p and all integers $\nu \geqslant 0$, we have

$$\lambda_f(p^\nu) = \alpha_f(p)^\nu + \alpha_f(p)^{\nu-1}\beta_f(p) + \cdots + \beta_f(p)^\nu \tag{1.2}$$

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where

$$\begin{cases} \alpha_f(p) = \varepsilon_f(p)p^{-1/2}, \ \ \beta_f(p) = 0 & \text{if } p \,\|\, N \\ |\alpha_f(p)| = \alpha_f(p)\beta_f(p) = 1 & \text{if } p \nmid N \end{cases} \tag{1.3}$$

and $\varepsilon_f(p) = \pm 1$. The relation $|\alpha_f(p)|$ follows from Deligne [4], and plainly we have

$$|\lambda_f(n)| \leqslant d(n) \tag{1.4}$$

for all $n \geqslant 1$, where $d(n) := \sum_{d|n} 1$ is the divisor function.

Throughout the paper, we write $F = \mathrm{sym}^2 f$ for any $f \in \mathrm{H}_k^*(N)$. Our concerned object $\lambda_F(n)$ can be depicted by the Dirichlet series representation of the symmetric square L-function associated to f. Define

$$L(s, F) := \prod_p \prod_{0 \leqslant j \leqslant 2} \left(1 - \frac{\alpha_f(p)^{2-j}\beta_f(p)^j}{p^s}\right)^{-1}$$

for $\Re e\, s > 1$, and write its Dirichlet series as

$$L(s, F) = \sum_{n=1}^{\infty} \frac{\lambda_F(n)}{n^s}.$$

Then $\lambda_F(n)$ is also real, multiplicative and

$$|\lambda_F(n)| \leqslant d_3(n) \tag{1.5}$$

for all integers $n \geqslant 1$, where $d_3(n) := \sum_{m|n} d(m)$. Furthermore we have

$$\lambda_F(p^\nu) = p^{-\nu} \qquad (p \mid N,\, \nu \geqslant 0) \tag{1.6}$$

and

$$\lambda_F(dm) = \lambda_F(d)\lambda_F(m) \qquad (d \mid N^\infty,\, m \geqslant 1). \tag{1.7}$$

These can be easily verified with (1.2), (1.3), (1.4) and the definition of $\lambda_F(n)$. Introduce

$$\Lambda(s, F) := N^s L_\infty(s, F) L(s, F)$$

with $L_\infty(s, F) = 2^{-s}\pi^{-3s/2}\Gamma(s/2)\Gamma(s+k-1)$. It is known that $\Lambda(s, F)$ is entire and bounded in vertical strips, and it satisfies the functional equation

$$\Lambda(s, F) = \varepsilon_F \Lambda(1 - s, F) \tag{1.8}$$

on $\mathbb{C}$, where $\varepsilon_F = \pm 1$.

In this paper we shall consider four problems related to the values of $\lambda_F(n)$ where $f \in \mathrm{H}_k^*(N)$.

1.1. **Positive and negative values over a positive density of primes.** The Sato-Tate conjecture asserts that for any $[u, v] \subset [-2, 2]$,

$$\sum_{\substack{p \leqslant x,\, p \nmid N \\ u \leqslant \lambda_f(p) \leqslant v}} 1 \sim \frac{x}{\log x} \int_u^v \frac{\sqrt{4 - t^2}}{2\pi}\, \mathrm{d}t \tag{1.9}$$

as $x \to \infty$. This conjecture has been proved for elliptic curves over $\mathbb{Q}$ with multiplicative reduction at some prime by Clozel, Harris & Taylor [3]. We also remark

that there is much ongoing progress in the general case for non-CM $f \in \mathrm{H}_k^*(N)$ (where N is not necessarily squarefree); in fact, a proof of the conjecture has been announced by Barnet-Lamb, Geraghty, Harris and Taylor [1, Theorem B]. The distribution of $\lambda_F(p)$ follows from Sato-Tate's conjecture, because by definition, we have

$$\lambda_F(n) = \sum_{d^2m=n} \lambda_f(m^2) \tag{1.10}$$

for $(n, N) = 1$, and in particular

$$\lambda_F(p) = \lambda_f(p^2) = \lambda_f(p)^2 - 1 \qquad (p \nmid N). \tag{1.11}$$

By (1.9), we infer that for any $[u, v] \subset [-1, 3]$,

$$\begin{aligned} \sum_{\substack{p\leqslant x,\, p\nmid N\\ u\leqslant \lambda_F(p)\leqslant v}} 1 &= \sum_{\substack{p\leqslant x,\, p\nmid N\\ \sqrt{1+u}\leqslant \lambda_f(p)\leqslant \sqrt{1+v}}} 1 + \sum_{\substack{p\leqslant x,\, p\nmid N\\ -\sqrt{1+v}\leqslant \lambda_f(p)\leqslant -\sqrt{1+u}}} 1 \\ &\sim \frac{2x}{\log x} \int_{\sqrt{1+u}}^{\sqrt{1+v}} \frac{\sqrt{4-t^2}}{2\pi}\,\mathrm{d}t \\ &\sim \frac{x}{\log x} \int_u^v \sqrt{\frac{3-t}{t+1}} \frac{\mathrm{d}t}{2\pi} \end{aligned} \tag{1.12}$$

as $x \to \infty$.

Here we re-tour Serre's elegant method in [16, Appendix] to derive some thresholds of positive and negative values exceeded by $\lambda_F(p)$ for a positive density of primes. Our result is certainly superseded by (1.12) on the basis of the (settled) Sato-Tate's conjecture. We would include below since it plays a role in the proof of Theorems 2 and 3.

Theorem 1. *Let $k \geqslant 2$ be an even integer, $N \geqslant 1$ be squarefree and $f \in \mathrm{H}_k^*(N)$.*

(i) *For any $\varepsilon > 0$, there is a positive density of primes p such that*

$$\lambda_F(p) \leqslant -5/7 + \varepsilon. \tag{1.13}$$

(ii) *For any $\varepsilon > 0$, there is a positive density of primes p such that*

$$\lambda_F(p) \geqslant 2 - \varepsilon. \tag{1.14}$$

Remark 1. The assertions (1.13) and (1.14) were known, see R. Murty [14, Corollaries 1 and 2], if we took the weaker constants $-1/3$ and 1 in place of $-5/7$ and 2. Subsequently the constant 1 was improved to $4\cos^2(2\pi/11) - 1 \approx 1.83$ by Kim & Shahidi [9, Proposition 4.11].

1.2. **$\Omega_\pm$-results over integers.** In virtue of (1.4) and the order of $d_3(n)$, we have

$$|\lambda_F(n)| \leqslant \exp\left\{(\log 4 + o(1))\frac{\log n}{\log_2 n}\right\} \tag{1.15}$$

for all integers $n \geqslant 2$, where $\log_r$ denotes the r-fold iterated logarithm.

The next theorem gives an $\Omega_\pm$-estimate showing that the factor in the exponent is of the right order of magnitude.

Theorem 2. *Let $k \geqslant 2$ be an even integer and $N \geqslant 1$ be squarefree. Then for each $f \in \mathrm{H}_k^*(N)$, there are two positive constants $C_\pm(f)$ such that*

$$\lambda_F(n) = \Omega_\pm\bigg(\exp\bigg\{ C_\pm(f) \frac{\log n}{\log_2 n} \bigg\} \bigg)$$

as $n \to \infty$.

In view of (1.15), the room for improvement in the result of Theorem 2 is the values of the constants $C_\pm(f)$.

1.3. **The number of same signs in long intervals.** Evidently (1.12) implies that the density of integers n where $\lambda_F(n)$ is positive (resp. negative) is positive. We shall give this consequence a proof using a number-theoretic method together with the fact that the density for $\lambda_F(p) = a$ is zero for each fixed a. This density result was already established in [15]. Define

$$\mathscr{N}_F^\pm(x) := \sum_{\substack{n \leqslant x,\, (n,N)=1 \\ \lambda_F(n) \gtrless 0}} 1. \tag{1.16}$$

Our result is as follows.

Theorem 3. *Let $k \geqslant 2$ be an even integer and $N \geqslant 1$ be squarefree. Then for any $f \in \mathrm{H}_k^*(N)$, we have*

$$\mathscr{N}_F^\pm(x) \gg_f x$$

for $x \geqslant x_0(f)$.

Actually our proof gives the stronger result

$$\sum_{\substack{n \leqslant x,\, (n,N)=1 \\ n \text{ squarefree},\, \lambda_F(n) \gtrless 0}} 1 \gg_f x \qquad (x \geqslant x_0(f)).$$

One can evaluate explicitly the value of $x_0(f)$ in term of N and k by applying an unpublished result of Kowalski [10, Proposition 1] instead of [15, Theorem 15] of Serre (see also (4.1) below). We hope to discuss this in another occasion.

1.4. **The number of same signs in short intervals.** Now we consider the sign problem over short intervals in order to visualize the fluctuation of $\lambda_F(n)$. Such a result cannot be deduced by (1.12) unless we know its rate of convergence.

Theorem 4. *Let $k \geqslant 2$ be an even integer, $N \geqslant 1$ be squarefree and $f \in \mathrm{H}_k^*(N)$. Set $\Psi(N) := \sum_{d|N} d^{-2/3}$ and $N_1 := N\Psi(N)^{3/4}$. There is an absolute constant $C > 0$ such that for any $\varepsilon > 0$ and all sufficiently large $x \geqslant N_1^2 x_0(k)$, we have*

$$\mathscr{N}_F^\pm(x + C(N_1 x)^{2/3}) - \mathscr{N}_F^\pm(x) \gg_\varepsilon (Nx)^{1/3-\varepsilon}, \tag{1.17}$$

where $x_0(k)$ is a suitably large constant depending on k and the implied constant in $\gg_\varepsilon$ depends only on ε.

Remark 2. We have $1 \leqslant \Psi(N) \leqslant \exp\{(3 + o(1))(\log N)^{1/3}/(\log_2 N)\}$.

The method of proof is the same as in [13, Theorem 2]. Thus the key turns to be an analogue of [13, Lemma 3.2], which is more delicate in the current situation because the O-term for the truncated Voronoi-type formula, derived using the method like

[13, Lemma 3.1], is $O(x^{1/3+\varepsilon})$. But our argument detect points attaining only values $\asymp x^{1/3}$, which cannot dominate the O-term. We shall get around this difficulty with the tools developed in [6] and [12] in the pursuit of the works by Chandrasekharan and Narasimhan [2].

2. Proof of Theorem 1

Let $T_n(x)$ be the polynomial which gives the trace of the nth symmetric power of an element of $SL_2(\mathbb{C})$ whose trace is x. For our purpose we list the explicit formula for $n \leqslant 8$:

$$\begin{aligned} T_0(x) &:= 1, \\ T_1(x) &:= x, \\ T_2(x) &:= x^2 - 1, \\ T_3(x) &:= x^3 - 2x, \\ T_4(x) &:= x^4 - 3x^2 + 1, \\ T_5(x) &:= x^5 - 4x^3 + 3x, \\ T_6(x) &:= x^6 - 5x^4 + 6x^2 - 1, \\ T_7(x) &:= x^7 - 6x^5 + 10x^3 - 4x, \\ T_8(x) &:= x^8 - 7x^6 + 15x^4 - 10x^2 + 1. \end{aligned} \tag{2.1}$$

It is well known that $T_0(x), T_1(x), \dots, T_8(x)$ constitutes a base of $V_8(x)$ where $V_d(x)$ denotes the vector space of all real polynomials of degree $\leqslant d$ over $\mathbb{R}$.

For any $\phi \in V_8(x)$, we define

$$I_f(\phi) := \lim_{x\to\infty} \frac{1}{\pi(x)} \sum_{p\leqslant x} \phi(\lambda_f(p)), \tag{2.2}$$

where $\pi(x)$ is the number of primes $p \leqslant x$ so

$$\pi(x) \sim \frac{x}{\log x} \qquad (x \to \infty)$$

by the prime number theorem. Using [9, Propositions 4.2-4.6], we get

$$I_f(T_0) = 1 \qquad \text{and} \qquad I_f(T_n) = 0 \quad (n = 1, 2, \dots, 8), \tag{2.3}$$

that guarantee the existence of the limit in (2.2).

From (2.1) and (2.3), we can deduce that

$$\begin{aligned} x^2 &= T_2(x) + T_0(x), \\ x^4 &= T_4(x) + 3T_2(x) + 2T_0(x), \\ x^6 &= T_6(x) + 5T_4(x) + 9T_2(x) + 5T_0(x), \\ x^8 &= T_8(x) + 7T_6(x) + 20T_4(x) + 28T_2(x) + 14T_0(x), \end{aligned}$$

and

$$I_f(1) = 1, \quad I_f(x^2) = 1, \quad I_f(x^4) = 2, \quad I_f(x^6) = 5, \quad I_f(x^8) = 14. \tag{2.4}$$

To prove the assertion (i), we consider the polynomial

$$\begin{aligned} P(x) &:= 7T_8(x) - 9T_6(x) + 6T_4(x) - 10T_2(x) \\ &= 7x^8 - 58x^6 + 156x^4 - 152x^2 + 32 \\ &= (7x^2 - 2)(x^2 - 2)^2(x^2 - 4). \end{aligned}$$

Suppose the set of p's with $|\lambda_f(p)| \leqslant \sqrt{2/7 + \varepsilon}$ has zero density. Then

$$\sum_{\substack{p \leqslant x \\ |\lambda_f(p)| < \sqrt{2/7+\varepsilon}}} 1 = o\big(\pi(x)\big) \tag{2.5}$$

as $x \to \infty$.

On the other hand, we have, for $2 \geqslant |x| \geqslant \sqrt{2/7 + \varepsilon}$,

$$\begin{aligned} P(x) &\leqslant 7\varepsilon(x^2 - 2)^2(x^2 - 4) \\ &= 7\varepsilon(x^6 - 8x^4 + 20x^2 - 16). \end{aligned}$$

Together with (2.5) and Deligne's inequality, it follows that

$$\begin{aligned} \sum_{p \leqslant x} P(\lambda_f(p)) &= \sum_{\substack{p \leqslant x \\ |\lambda_f(p)| \geqslant \sqrt{2/7+\varepsilon}}} P(\lambda_f(p)) + o\big(\pi(x)\big) \\ &\leqslant 7\varepsilon \sum_{\substack{p \leqslant x \\ |\lambda_f(p)| \geqslant \sqrt{2/7+\varepsilon}}} \big(\lambda_f(p)^6 - 8\lambda_f(p)^4 + 20\lambda_f(p)^2 - 16\big) + o\big(\pi(x)\big) \\ &= 7\varepsilon \sum_{p \leqslant x} \big(\lambda_f(p)^6 - 8\lambda_f(p)^4 + 20\lambda_f(p)^2 - 16\big) + o\big(\pi(x)\big) \end{aligned}$$

as $x \to \infty$. Invoking (2.4), we find that

$$\sum_{p \leqslant x} P(\lambda_f(p)) \leqslant -49\varepsilon\pi(x) + o\big(\pi(x)\big) \tag{2.6}$$

as $x \to \infty$.

Combining (2.5) and (2.6) leads to the inequality

$$I_f(P) \leqslant -49\varepsilon.$$

However, as $P = 7T_8 - 9T_6 + 6T_4 - 10T_2$, (2.3) implies

$$I_f(P) = 0.$$

The contradiction shows a positive density of primes p at which $|\lambda_f(p)| \leqslant \sqrt{2/7 + \varepsilon}$. This concludes, via (1.11), the assertion (i).

The assertion (ii) can be attained in a similar way. Consider the polynomial

$$\begin{aligned} Q(x) &:= T_8(x) + 2T_6(x) + 2T_4(x) + T_2(x) \\ &= x^8 - 5x^6 + 7x^4 - 3x^2 \\ &= x^2(x^2 - 1)^2(x^2 - 3). \end{aligned}$$

If the set of p's with $|\lambda_f(p)| > \sqrt{3-\varepsilon}$ has zero density, then

$$\sum_{\substack{p\leqslant x\\ |\lambda_f(p)|>\sqrt{3-\varepsilon}}} 1 = o\big(\pi(x)\big) \tag{2.7}$$

as $x\to\infty$. For $|x|\leqslant\sqrt{3-\varepsilon}$, we have

$$Q(x)\leqslant -\varepsilon x^2(x^2-1)^2 = -\varepsilon(x^6-2x^4+x^2),$$

whence it follows

$$\begin{aligned}\sum_{p\leqslant x} Q(\lambda_f(p)) &= \sum_{\substack{p\leqslant x\\ |\lambda_f(p)|\leqslant\sqrt{3-\varepsilon}}} Q(\lambda_f(p)) + o\big(\pi(x)\big)\\ &\leqslant -\varepsilon \sum_{\substack{p\leqslant x\\ |\lambda_f(p)|\leqslant\sqrt{3-\varepsilon}}} \big(\lambda_f(p)^6 - 2\lambda_f(p)^4 + \lambda_f(p)^2\big) + o\big(\pi(x)\big)\\ &= -\varepsilon\sum_{p\leqslant x} \big(\lambda_f(p)^6 - 2\lambda_f(p)^4 + \lambda_f(p)^2\big) + o\big(\pi(x)\big)\end{aligned}$$

as $x\to\infty$.

Using (2.4), we infer that

$$\sum_{p\leqslant x} Q(\lambda_f(p)) \leqslant -2\varepsilon\pi(x) + o\big(\pi(x)\big)$$

as $x\to\infty$ and thus, with (2.7),

$$I_f(Q)\leqslant -2\varepsilon.$$

This contradicts to

$$I_f(Q) = 0,$$

evaluated by (2.3) and the definition of $Q(x)$. The assertion (ii) is thus complete with (1.11).

3. Proof of Theorem 2

Immediately from Theorem 1(ii), the set of primes

$$\mathscr{P}_f := \big\{p : p\nmid N \text{ and } \lambda_F(p)\geqslant 1.9\big\}$$

has a positive lower density. Thus there are two positive constants $C_1(f)$ and $x_0(f)$ such that

$$\sum_{p\leqslant x,\, p\in\mathscr{P}_f} 1 \geqslant C_1(f)\frac{x}{\log x} \tag{3.1}$$

for $x\geqslant x_0(f)$.

Define

$$m_j := \prod_{p\leqslant j,\, p\in\mathscr{P}_f} p.$$

Then

$$\lambda_F(m_j) = \prod_{p\leqslant j,\, p\in\mathscr{P}_f} \lambda_F(p) \geqslant 1.9^{C_1(f)j/\log j}$$

provided $j \geqslant x_0(f)$.

Observing that

$$\log m_j = \sum_{p\leqslant j,\, p\in\mathscr{P}_f} \log p \leqslant 10j$$

for all integers $j \geqslant 1$, we deduce with the preceding inequality that

$$\lambda_F(m_j) \geqslant 1.9^{C_2(f)\log m_j/\log_2 m_j}$$

for some positive constant $C_2(f)$, provided $j \geqslant x_0(f)$. This implies the required Ω_+-result with $C_+(f) = C_2(f)\log 1.9$.

To get the Ω_--result, we fix a prime p' such that $\lambda_F(p') \leqslant -5/7$, whose existence is justified by Theorem 1(i). Setting $n_j := p'm_j$, we have

$$\begin{aligned}\lambda_F(n_j) &= \lambda_F(p')\lambda_F(m_j)\\ &\leqslant -\frac{5}{7}\exp\left\{C_+(f)\frac{\log m_j}{\log_2 m_j}\right\}\\ &\leqslant -\frac{5}{7}\exp\left\{\frac{C_+(f)}{2}\frac{\log n_j}{\log_2 n_j}\right\}\end{aligned}$$

for $j \geqslant x_0(f,p')$. This completes the proof.

4. Proof of Theorem 3

Let p' be the least prime such that $p' \nmid N$ and $\lambda_F(p') = \lambda_f(p'^2) \leqslant -5/7$ (see Theorem 1 (i) for its existence). Define

$$\mathscr{P}_{f,a} := \{p \in \mathscr{P} : p \nmid N \text{ and } \lambda_f(p) = a\}$$

where $\mathscr{P}$ is the set of all prime numbers.

Serre proved in [15, Theorem 15] that

$$\big|[1,x]\cap\mathscr{P}_{f,a}\big| \ll_{f,\delta,a} \frac{x}{(\log x)^{1+\delta}} \tag{4.1}$$

for all $x \geqslant 2$, and for any fixed $\delta < \frac{1}{2}$ and any fixed $a \in \mathbb{R}$.

Write

$$\mathscr{P}' := \mathscr{P}_{f,-1} \cup \mathscr{P}_{f,1} \cup \{p \in \mathscr{P} : p \mid p'N\}$$

and

$$\begin{aligned}\mathscr{B} &= \mathscr{P}' \cup \big\{p^2 : p \in \mathscr{P}\smallsetminus\mathscr{P}'\big\}\\ &= \{b_i\}_{i\geqslant 1} \quad \text{(with increasing order)}.\end{aligned}$$

In light of (4.1), we have

$$\sum_{i\geqslant 1}\frac{1}{b_i} < \infty \qquad \text{and} \qquad (b_i,b_j) = 1 \quad (i\neq j).$$

Then we consider the sequence of all $\mathscr{B}$-free numbers, $\mathscr{A} := \{a_i\}_{i\geqslant 1}$ (with increasing order). That is, the a_i's in $\mathscr{A}$ are all positive integers indivisible by any element in

$\mathscr{B}$. In [5], Erdös showed that $\mathscr{A}$ is of density

$$\lim_{x\to\infty}\frac{|\mathscr{A}\cap[1,x]|}{x}=\prod_{i=1}^{\infty}\left(1-\frac{1}{b_i}\right)>0. \tag{4.2}$$

Clearly each $a\in\mathscr{A}$ is squarefree. For all $a\in\mathscr{A}$ we have

$$\lambda_F(a)=\prod_{p|a}\lambda_F(p)=\prod_{p|a}\lambda_f(p^2)=\prod_{p|a}\big(\lambda_f(p)^2-1\big)\neq 0$$

using the multiplicativity of $\lambda_f(n)$ and the Hecke relation. Next we partition

$$\mathscr{A}=\mathscr{A}^+\cup\mathscr{A}^-,$$

where

$$\mathscr{A}^{\pm}:=\big\{a_i\in\mathscr{A}:\lambda_F(a_i)\gtrless 0\big\},$$

then the switched sets

$$\mathscr{N}^{\pm}:=\mathscr{A}^{\pm}\cup\{a_ip':a_i\in\mathscr{A}^{\mp}\}$$

satisfy respectively $\lambda_F(a)\gtrless 0$ for all $a\in\mathscr{N}^{\pm}$. Note that $(a,p')=1$ for $a\in\mathscr{A}^{\pm}$. Therefore,

$$\mathscr{N}_F^{\pm}(x)\geqslant\big|\mathscr{N}^{\pm}\cap[1,x]\big|\geqslant\big|\mathscr{A}\cap[1,x/p']\big|$$

for all $x\geqslant 1$. The desired result follows with the inequality (4.2).

5. Proof of Theorem 4

Parallel to [13, Lemma 3.2], we establish the following lemma but as explained in Section 1, the proof there does not apply. Concerning with the Voronoi formula which was firstly introduced to the error term in the Dirichlet divisor problem, it was greatly extended to a general context by Chandrasekharan and Narasimhan and further studied by a number of people. Simply speaking, the Voronoi-type formula was established for a class of error terms generated by (generalized) Dirichlet series which satisfy a functional equation of Riemann-type (as in Selberg class). However there is an intricate problem of convergence when the degree gets higher. This theory is applicable here in light of the analytic properties of $\Lambda(s,F)$ and the functional equation (1.8).

Lemma 5.1. *Let $f\in\mathrm{H}_k^*(N)$ and N_1 be defined as in Theorem 4. There exist positive absolute constants C,c_1,c_2 such that for all sufficiently large $X\geqslant N_1^2X_0(k)$, we can find $x_1,x_2\in[X,X+C(N_1X)^{2/3}]$ for which*

$$S_F^*(x_1)>c_1(NX)^{1/3}\qquad\text{and}\qquad S_F^*(x_2)<-c_2(NX)^{1/3},$$

where $X_0(k)$ is a constant depending only on k and

$$S_F^*(x):=\sideset{}{'}\sum_{\substack{n\leqslant x\\(n,N)=1}}\lambda_F(n).$$

Here the dash is to indicate that the last summand is halved when $x\in\mathbb{N}$.

Proof. We begin the proof with the results in [6] and [12]. More specifically we adopt the notation of [12] and set $a_n = \lambda_F(n)$, $b_n = \varepsilon_F a_n$, $\lambda_n = \mu_n = 2\pi^{3/2}n/N$, so

$$\phi(s) = 2^{-s}\pi^{-3s/2}N^s L(s,F), \qquad \psi(s) = \varepsilon_F\phi(s), \qquad \sigma_a^* = \sigma_b^* = 1.$$

Moreover $\Delta(s) = L_\infty(s,F)$ and $\delta = 1$, thus $\alpha_1 = \frac{1}{2}$, $\beta_1 = 0$ and $\alpha_2 = 1$, $\beta_2 = k-1$. With some calculation, one sees that the symbols in [12, Lemma 2.2] take the values

$$\alpha = \frac{3}{2}, \quad h = 3\cdot 2^{1/3}, \quad \theta_\rho = \frac{1+2\rho}{3}, \quad k_\nu(\rho) = -k - \frac{\nu-\rho}{2} \quad (\nu = 0,1),$$
$$e_0(\rho) = (3\pi)^{-1/2}2^{(1-\rho)/3}, \quad e_1(\rho) = e_0(\rho)Q_2(\rho)$$

for some quadratic polynomial Q_2. Remark that a factor of $\sqrt{2}$ is missing in the formula for $e_0(\rho)$ in [12] (and [6] too), i.e. the correction is $e_0(\rho) = \sqrt{2}(3/h)^\rho/\sqrt{h\pi}$.

Let $\rho \in (0,1]$ be any number and define

$$S_F(x,\rho) := \frac{1}{\Gamma(\rho+1)}\sideset{}{'}\sum_{n\leqslant x}\lambda_F(n)(x-n)^\rho,$$

where the dash is to indicate that the last summand is halved when $\rho = 0$ and $x \in \mathbb{N}$. By [12, Lemmas 2.1 and 2.2] with $2\pi^{3/2}x/N$ in place of x, we infer that

$$S_F(x,\rho) = \varepsilon_F\biggl(\frac{N}{2\pi^{3/2}}\biggr)^{1+2\rho}\sum_{n\geqslant 1}\frac{\lambda_F(n)}{n^{1+\rho}}G_\rho\biggl(\frac{4\pi^3 nx}{N^2}\biggr) \tag{5.1}$$

where

$$\begin{aligned} G_\rho(y) = {} & \frac{(2^{1-\rho}y^{1+2\rho})^{1/3}}{\sqrt{3\pi}}\cos\biggl(3(2y)^{1/3} - \Bigl(k+\frac{\rho}{2}\Bigr)\pi\biggr) \\ & + e_1(\rho)y^{2\rho/3}\cos\biggl(3(2y)^{1/3} - \Bigl(k+\frac{\rho-1}{2}\Bigr)\pi\biggr) + O\biggl(\frac{1}{y^{(1-2\rho)/3}}\biggr) \end{aligned} \tag{5.2}$$

and the O-constant is uniform for $\rho \in [0,1]$. The subtle point here is that the right-handed side of (5.1) is ensured to be a uniformly convergent series on any finite closed interval in $(0,\infty)$ only under the condition $\rho > 0$, although its left-handed side is well-defined for $\rho \geqslant 0$. Hence we cannot immediately apply (5.1) with $\rho = 0$ and need a process of taking limit to circumvent the prohibition.

Let

$$\Phi(v,\rho) := (2\pi)^\rho\pi\sqrt{3}\frac{S_F(N^2v^3,\rho)}{(Nv)^{1+2\rho}} \qquad \text{and} \qquad \Phi_d(v,\rho) := \Phi\biggl(\frac{v}{d^{1/3}},\rho\biggr)$$

where $1 \leqslant d \leqslant N$. Then by (5.1) and (5.2),

$$\begin{aligned} \Phi(v,\rho) = {} & \varepsilon_F\sum_{n\geqslant 1}\frac{\lambda_F(n)}{n^{(2+\rho)/3}}\cos\biggl(6\pi n^{1/3}v - \frac{\rho}{2}\pi\biggr) \\ & - \frac{\varepsilon_F}{(4\pi^3)^{1/3}v}\sum_{n\geqslant 1}\frac{\lambda_F(n)}{n^{1+\rho/3}}\sin\biggl(6\pi n^{1/3}v - \frac{\rho}{2}\pi\biggr) + O\biggl(\frac{1}{v^2}\biggr). \end{aligned} \tag{5.3}$$

We emphasize again the O-constant here is independent of $\rho \in (0,1]$.

Let t be any large natural number, $\kappa > 1$ a large parameter taken at our disposal and write

$$K_\tau(u) = (1-|u|)(1+\tau\cos(6\pi\kappa u))$$

with $\tau = \pm 1$. We consider the integral

$$J_{\tau,d}(\rho) = \int_{-1}^{1} \Phi_d(t+\kappa u, \rho) K_\tau(u)\,\mathrm{d}u. \tag{5.4}$$

In light of (5.3), we are led to evaluate the integrals

$$r_\beta(\rho) := \int_{-1}^{1} K_\tau(u) \cos\left(6\pi\beta(t+\kappa u) - \frac{\rho}{2}\pi\right)\mathrm{d}u$$

and

$$s_\beta(\rho) := \int_{-1}^{1} \frac{K_\tau(u)}{t+\kappa u} \sin\left(6\pi\beta(t+\kappa u) - \frac{\rho}{2}\pi\right)\mathrm{d}u$$

where $\beta = (n/d)^{1/3}$ ($n \in \mathbb{N}$) in practice. We remark that the condition $\rho > 0$ is applied to justify the legitimacy in the interchange of the integration and summations.

As in [13, (3.13)] and [7, (2.3)], we have

$$r_\beta(\rho) = \delta_{\beta=1}\frac{\tau}{2}\cos\left(\frac{\rho}{2}\pi\right) + O\bigg(\min\left\{1, \frac{1}{\kappa^2\beta^2}\right\} + \delta_{\beta\neq 1}\min\left\{1, \frac{1}{\kappa^2(\beta-1)^2}\right\}\bigg)$$

and

$$s_\beta(\rho) \ll (t\beta\kappa)^{-1}$$

where the implied constants in the O-symbols and $\ll$-symbol are independent of ρ. After inserting (5.3) into (5.4), the O-constant in (5.3) gives $O(d^{2/3}t^{-2})$ while the infinite sum of the second summand is

$$\ll \frac{d^{1/3}}{\kappa t}\sum_{n\geqslant 1}\frac{|\lambda_F(n)|}{n^{4/3+\rho}} \ll \frac{d^{1/3}}{\kappa t}.$$

Therefore,

$$J_{\tau,d}(\rho) = \frac{\tau}{2}\cos\left(\frac{\rho}{2}\pi\right) + O\bigg(\frac{d^{1/3}}{\kappa t} + \frac{d^{2/3}}{t^2} + R\bigg), \tag{5.5}$$

where

$$R := \sum_{n\geqslant 1}\frac{|\lambda_F(n)|}{n^{(2+\rho)/3}}\min\left\{1, \frac{d^{2/3}}{\kappa^2 n^{2/3}}\right\} + \sum_{\substack{n\geqslant 1\\ n\neq d}}\frac{|\lambda_F(n)|}{n^{(2+\rho)/3}}\min\left\{1, \frac{d^{2/3}}{\kappa^2(n^{1/3}-d^{1/3})^2}\right\}$$

and the O-constant is uniform for $\rho \in (0,1]$. Following the treatment for the sums in [13, (3.15)], we obtain $R \ll (d/\kappa)^{1/3}\log^2(2d)$, where the implied constant is absolute. It follows that

$$J_{\tau,d}(\rho) = \frac{\tau}{2}\cos\left(\frac{\rho}{2}\pi\right) + O\bigg(\frac{d^{1/3}\log^2(2d)}{\kappa^{1/3}} + \frac{d^{2/3}}{t^2}\bigg)$$

holds for all $\rho \in (0,1]$, where we have removed the term $d^{1/3}/(\kappa t)$ since

$$\frac{d^{1/3}}{\kappa t} \leqslant \frac{d^{1/2}\log(2d)}{\kappa^{1/6}t} = \bigg(\frac{d^{1/3}\log^2(2d)}{\kappa^{1/3}}\bigg)^{1/2}\bigg(\frac{d^{2/3}}{t^2}\bigg)^{1/2}.$$

Taking $\rho \to 0+$, we conclude

$$J_{\tau,d}(0) = \frac{\tau}{2} + O\bigg(\frac{d^{1/3}\log^2(2d)}{\kappa^{1/3}} + \frac{d^{2/3}}{t^2}\bigg). \tag{5.6}$$

In view of (1.7), the Möbius inversion formula allows us to deduce

$$S_F^*(x) = \sum_{d|N} \mu(d)\lambda_F(d) S_F\Big(\frac{x}{d}, 0\Big),$$

where $\mu(n)$ is the Möbius function. Write

$$J_\tau := \frac{\pi\sqrt{3}}{\varepsilon_F} \int_{-1}^{1} \frac{S_F^*(N^2(t+\kappa u)^3)}{N(t+\kappa u)} K_\tau(u)\,\mathrm{d}u,$$

then by (5.4) and (5.6),

$$\begin{aligned} J_\tau &= \sum_{d|N} \frac{\mu(d)}{d^{1/3}} \lambda_F(d) J_{\tau,d}(0) \\ &= \frac{\tau}{2} \sum_{d|N} \frac{\mu(d)}{d^{4/3}} + O\bigg(\frac{1}{\kappa^{1/3}} \sum_{d|N} \frac{\log^2(2d)}{d} + \frac{1}{t^2} \sum_{d|N} \frac{1}{d^{2/3}} \bigg) \\ &= \frac{\tau}{2} \prod_{p|N} \Big(1 - \frac{1}{p^{4/3}}\Big) + O\bigg(\frac{\Psi(N)^{1/6}}{\kappa^{1/3}} + \frac{\Psi(N)}{t^2} \bigg) \end{aligned}$$

for

$$\begin{aligned} \sum_{d|N} \frac{\log^2(2d)}{d} &\leqslant \bigg\{ \sum_{d|N} (d^{-1/9})^6 \bigg\}^{1/6} \bigg\{ \sum_{d|N} \bigg(\frac{\log^2(2d)}{d^{8/9}} \bigg)^{6/5} \bigg\}^{5/6} \\ &\ll \Psi(N)^{1/6}. \end{aligned}$$

Let $C > 0$ be a large constant. When $X \geqslant C^3 N_1^2$, we take $t = [(X/N^2)^{1/3}]$ and $\kappa = \frac{1}{4} C\sqrt{\Psi(N)}$, so $t > 2\kappa$. Note that $\prod_{p|N}(1 - p^{-4/3}) > \zeta(4/3)^{-1} \gg 1$ and the O-term in J_τ is $\ll C^{-1/3}$, so the main term dominates when C is suitably large. Therefore

$$J_{-1} < -\frac{1}{4\zeta(4/3)} \qquad \text{and} \qquad J_1 > \frac{1}{4\zeta(4/3)}.$$

The kernel function $K_\tau(u)$ is nonnegative and satisfies

$$1 - (3\pi\kappa)^{-2} \leqslant \int_{-1}^{1} K_\tau(u)\,\mathrm{d}u \leqslant 2 \qquad (\tau = \pm 1).$$

Thus we have

$$2\pi\sqrt{3} \frac{S_F^*(N^2(t+\kappa\eta_+)^3)}{N(t+\kappa\eta_+)} \geqslant \frac{1}{4\zeta(4/3)}$$

and

$$\big(1 - (3\pi\kappa)^{-2}\big)\pi\sqrt{3} \frac{S_F^*(N^2(t+\kappa\eta_-)^3)}{N(t+\kappa\eta_-)} \leqslant -\frac{1}{4\zeta(4/3)}$$

for some $\eta_\pm \in [-1, 1]$. These two points deviate from X by a distance $\ll (NX)^{1/3}$, since the difference between $N^2(t \pm \kappa)^3$ is $\ll \kappa N^2 t^2 \asymp (N_1 X)^{2/3}$ and $Nt \asymp (NX)^{1/3}$. This implies the result of Lemma 5.1. □

Now we proceed with the argument in [13] to prove Theorem 4.

By Lemma 5.1, for any $x \geqslant N_1^2 X_0(k)$ we can pick three points $x < x_1 < x_2 < x_3 < x + 3C(N_1x)^{2/3}$ such that $S_F^*(x_i) < -c(Nx)^{1/3}$ $(i = 1, 3)$ and $S_F^*(x_2) > c(Nx)^{1/3}$ for some absolute constant $c > 0$. (Note that $y + C(N_1y)^{2/3} \leqslant x + 3C(N_1x)^{2/3}$ for $y = x + C(N_1x)^{2/3}$.) Hence we deduce that

$$\sum_{\substack{x_1<n<x_2\\ \lambda_F(n)>0}} \lambda_F(n) \geqslant S_F^*(x_2) - S_F^*(x_1) > 2c(Nx)^{1/3}$$

and

$$\sum_{\substack{x_2<n<x_3\\ \lambda_F(n)<0}} \big(-\lambda_F(n)\big) \geqslant -\big(S_F^*(x_3) - S_F^*(x_2)\big) > 2c(Nx)^{1/3}.$$

Thus, Theorem 4 follows as each term in the two sums are positive and $\ll x^{\varepsilon}$.

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